

SECTION - A (MCQ)

1. Choose the correct answer.

- i. The first step in the Scientific Process is
Hypothesis - Observation - Experiment
- ii. Fossils are mostly found in rocks
Igneous rocks - Sedimentary rocks - Metamorphic rocks
- iii. The disease caused by bacteria is
Cholera - Cancer - Aids
- iv. The SI unit of force is
Metre - Newton - Kg
- v. The biggest Planet in our Solar System is
Jupiter - Mercury - Venus
- vi. Kinetic energy is due to
Rest - Motion - Height
- vii. Metamorphic rocks have the example of
Marble - Coal - Granite
- viii. Changing of water into ice is a
Physical Change - Chemical Change - Chemical Change
- ix. our Earth is a
galaxy - Star - Planet
- x. Solution is a
Homogenous mixture - Heterogenous mixture



Government College of Education Karachi

INTERNAL B* AREA BLOCK AS

B. Ed. Hons (4 Year Program), 3rd SEMESTER

MID TERM EXAMINATION - 2018
EARTH SCIENCE

Max. Marks: 20

Time Allowed: Two Hours

Instructions: Q. #1 is compulsory. Attempt any five questions from Section B and two from Section C.

(Marks- 05)

SECTION A

Q1. Encircle the correct answer

1. Einstein's famous equation is _____
a) $E=mc^2$ b) $E=mg$ c) $E=mv^2$ d) $E=mc$
2. The unit of energy is _____ in MKS system
a) gm b) cm c) joule d) m
3. Complete equipment of electrolysis is known as _____
a) Electrolysis process b) Electrolytic process c) Electrolytic current d) Electrolytic cell
4. An atom with charge is called _____
a) Molecule b) metal c) anion d) electron
5. _____ was awarded noble prize in physics.
a) Dr. Ahsan Raza b) Dr. Abdul Salam c) Dr. Samar Mubarak d) Dr. Abdul Qader
6. The application for science especially for industrial or commercial objectives is called _____
a) Technology b) Electronics c) Programming d) Radiation
7. A substance that burned to produce heat and light is known as _____
a) Alcohol b) Fuel c) Acid d) Alkaline
8. When water is solvent the solution is called _____
a) Saturated b) Aqueous c) Unsaturated d) None of them
9. A part of solution which is present in less quantity is called _____
a) Solution b) Solvent c) Solute d) None of them
10. Jabir Bin Hayan is known as the father of _____
a) Biology b) Chemistry c) Physics d) None of them

SECTION-B (SHORT Q/A)

(Marks-05)

1. What is Science?
2. What is meant by Bond Energy?
3. Write three characteristics of Covalent Bond.
4. What is the difference between exothermic and endothermic reaction?
5. What is energy? How many types of energy? Name them.
6. How many types of Solution?
7. Write three points about the positive impact of science in our life.

SECTION-C (LONG Q/A)

(Marks-10)

1. Define the law of Conservation of Mass with Examples.
2. Differentiate the following:
a) Kinetic Energy and Potential Energy
b) Chemical Reaction and Physical Reaction
3. What is Electrolysis? What are the uses of Electrolysis in our life?
4. What is chemical bond? How many types of chemical bond? Explain with suitable examples.
5. Write the names of Ten Muslim Scientists? Explain any three of them in detail.
6. What is meant by scientific Method? Also explain its different steps.
7. What is chemical reaction? How many types of chemical Reaction? Explain.

Q2. Give short answers of any five (5) of the following:

- i. ✓ Describe the nature of science.
- ii. ✓ What is meant by Radiation?
- iii. ✓ What type of visual aid is commonly used in Pakistan's educational institutions? And why?
- iv. ✓ Explain the law of conservation of mass.
- v. ✓ What is the importance of questioning in science teaching?
- vi. ✓ What is scientific method? Also write its different steps.
- vii. ✓ Write four lines of positive impact of science in our life.
- viii. ✓ Define the following

✓ (A) Energy

✓ (A) Solubility

✓ (C) Bond and its types (D) Lesson

Section 'C'

Q2. Give answers of any two (2) in detail:

- a. ✓ What is Electrolysis? Explain it with an experiment.
 - b. ✓ Make a lesson plan on any of the following topics.
(A) Energy and its types (B) Parts of flower.
 - c. ✓ What is chemical change? Also describe its various types.
- OR
- d. ✓ Write the name of five Muslim and five Non-Muslim Scientists. Also explain the contribution of two Muslim Scientists in the field of Biology.
 - e. ✓ Prepare an activity at elementary level for a lesson from the given topics.
(A) Solar System (B) Germination of Seeds

OR

Differentiate the following:

- i. ✓ Observation and Inference
- ii. ✓ Conduction and convection
- iii. ✓ Aims and objective
- iv. ✓ Chemical physical change
- v. ✓ Solute and solvent



Department of Teacher Education

University of Karachi

ADE/B Ed Hons (Elementary Program) External Exams Sep 2021

DTE-441-Teaching of General Science

Duration: 02 hrs.

(50+10 = 60)

"SECTION A"

Multiple Choice Questions

(10 Marks)

- 1) "Why do you burn your hands when you touch a hot plate?"
Which of the given phenomena is aligned with the question given above?
- ☒ a) Conduction b) Convection c) Radiation d) Transformation
- 2) "Friction on a moving object causes the object and the surface to warm up." This is an example of:
- ☒ a) Energy Transfer b) Energy Transformation c) Convection d) Conduction
- 3) All the following are examples of homogenous mixtures EXCEPT
- a) Alloy b) Soft drink c) Alcohol and water ☒ d) Oil and water
- 4) The ability of one substance to mix with another substance is known as
- ☒ (a) Chemical change. (b) Physical change.
(c) Solubility. (d) Electrolysis.
- 5) The reactions/chemical processes in which heat is absorbed are known as:
- a) Exothermic reactions. ☒ b) Endothermic reactions.
c) Electrolysis. d) Phase changes.

SECTION B

(Constructed Response Questions)

Marks: 50

Q1) Write a short note on any two of the following.

(20 marks)

- a) Inquiry based teaching approach
☒ b) Demonstration Method for teaching Science.
☒ c) Project based learning

Q2) Suggest a suitable activity for teaching "Chemical change" (OR) "Types of mixtures" through experiential learning strategy. Mention all the steps involved in the activity. (15 marks)

Q3) Design a lesson plan for teaching "Light and Shadow" to Grade One (OR) "Introduction to energy" to Grade Five students. (15 marks)

Section "B" (Short Answer Questions)

(marks: 20)

Attempt any 5 questions from this section. Each question carries 4 marks.

Q2 (i) . Define the following with examples of each:

- a) Solute b) physical changes c) exothermic reaction d) climate

(ii) Differentiate between the following:

- a) Kinetic and potential energy
b) Conduction and convection

(iii) Define reflection and state and explain its laws.

(iv) Give examples of different type of solutions on the basis of different states of matter.

(v) Give importance of teaching "human body as a system" at Elementary Level.

(vi) Design an activity to teach the topic "electrolysis" at Elementary level.

(vii) State and explain law of conservation of mass.

SECTION "C" (DETAILED ANSWER QUESTIONS) (marks: 30)

Note: attempt any three questions from this section. Each question carries 10 marks

Q3 (i) Define nature of science and its application in teaching at elementary level.

(ii) Define solubility and explain different factors affecting the solubility of gases and liquids.

(iii) Explain in detail the human circulatory system. (diagram is not required).

(iv) Write a note on any two of the following:

- a) Inquiry base teaching approach
b) Carbon cycle
c) Theory of tectonic plates

(marks: 10)

(marks: 10)

- (marks: 10)